



CAREER SUMMARY

- Experienced Data Scientist/AI Engineer with a proven track record of 8 years in ML, NLP, CV, LLMs, GenAI and Agent-based systems across Finance, FinTech and Operations domain.
- Certified Scrum Product Owner specialising in AI/ML, delivering solutions from POCs to MVPs with structured change management process and agile delivery frameworks.
- Skilled in stakeholder management for smooth delivery together with cross-functional teams.

PROFESSIONAL EXPERIENCE

OKX Singapore

Jul. 2024 - Dec. 2025

Senior Data Scientist, AI Science

- Implemented NLP, GenAI, LLM and ML solutions on cloud-native platforms (AWS, Alibaba PAI, Azure AI Foundry, Apache Kafka) to accelerate content growth, enhance platform retention, and support new user acquisition across OKX ecosystem. Delivered key initiatives in in-app content optimization, external content marketing, and foundational content intelligence for OKX Feed, OKX Assistant, AIGC & SEO Trending Articles and AI-based Localization & Translation.

Key Contributions

- Content-Based Capability Building
 - Fine-tuned transformer and sLLM-based models (LLaMA, Gemma, Phi etc.) for OKX Assistant and OKX Feed, enabling high-accuracy intent classification, token tagging, and tweet-level information extraction. Built LLM-based retraining workflows with automated data labelling and evaluation pipelines, enabling continuous improvement of model performance. Achieved > 90% accuracy in content deduplication through LLM-assisted matching and validation.
 - Enhanced contextual understanding for OKX Builder (a KOC-based UGC platform), by building a tokenomics-aware vector database aggregating multi-source crypto content. Deployed RAG systems and implemented MCP enhancements to improve query comprehension by LLMs.
 - Trained and deployed in-house multilingual translation models covering 20+ local OKX languages (<30% TER and >80% BLEU), reducing reliance on third-party services and cutting translation costs by 25%. Integrated ReAct-style LLM agents with Azure services to support a hybrid fallback translation strategy.
- Platform Content Growth & Retention
 - Delivered full-stack AIGC article generation pipelines for SEO-optimized, trend-aligned content creation (articles and images). Combined real-time hot topic detection with adaptive LLM-driven topic identification using models like GPT, Qwen, DeepSeek, and Claude. Extended with agentic orchestration (LangGraph, N8N) and integrated visual generation via DALL-E, coupled with real-time engagement monitoring for continuous optimization.
 - Delivered a GenAI-powered trader insight system that analyzed KOL tweets to extract and rank token signals (bullish/bearish) across 50+ tokens, refreshed hourly.
 - Fine-tuned 12B Flux.1 text-to-image models using branded assets (i.e., OKX logos) to support scalable visual generation for product and marketing use cases.
- Risk Mitigation
 - Developed ML-based models for high-value referral detection and high-risk user profiling using both behavioral and statistical indicators.
 - Built real-time on-chain transaction monitoring and malicious address detection systems using low-latency Redis streaming architectures.
- Internal Productivity Tools
 - Developed NLP and GenAI pipelines for Jira ticket classification and OEX Feedback Channels, enabling intelligent routing and efficient query resolution.

DBS Singapore

Jan. 2020 - Jun.2024

Senior Data Scientist, MOT Risk Department

- Assumed role of Product Owner, Data Scientist, ML Engineer within team to implement solutions to enhance the overall risk management process and creation of product roadmaps for AI/ML projects.
- Risk Management Analytics
 - Improved debt collection prioritization by 60% through customer segmentation and borrower risk scoring, leveraging K-Means, AWS Transcribe, PySpark, LLMs, and LangChain to extract actionable insights from call recordings.
 - Identified potential credit card delinquency risks within merchant networks using behavioral analytics, transaction pattern analysis, graph analytics, and peer analysis on large-scale datasets with PySpark, Neo4J, and anomaly detection algorithms.
 - Detected irregular borrowing behavior in the Hong Kong money-lending sector via analysis of round-trip transaction sand anomalies, using Neo4J network analytics and machine learning–based anomaly detection.
- NLP & MLOps Automation
 - Developed AAFA, a financial analysis automation tool, integrating NLP, GenAI, and MLOps to auto-generate corporate risk reports for Relationship Managers, achieving S\$900K cost savings. Applied OpenCV, financial NER, BERT classification, LLM summarization, spaCy extraction, and Evidently AI for drift detection; managed models via MLflow. Earned Hackett Digital Award 2022 and MOT People Award.
 - Led multiple GenAI proof-of-concepts (POCs), including contrarian surveillance, ESG compliance automation, and LLM-based call log analysis using LangChain, RAG, vector databases, and prompt engineering frameworks.
- ML Engineering
 - Integrated models into enterprise-grade APIs following Agile, CI/CD, and DevOps best practices using Jira, Jenkins, and Git.
 - Executed model training and deployment using Bitfusion GPUs, built PySpark jobs for large-scale data ingestion, and optimized HiveQL pipelines for efficient querying of distributed datasets.

AI Singapore

Dec. 2018 - Jan. 2020

AI Engineer, AI Technology department

- Assumed a key role in curating datasets, extensive research and scalable experimentations tailored towards AI Challenges' requirements.
- Engineered a scholarly knowledge graph in Neo4j using big data hosted on Azure, connecting papers, authors, institutions, and topics from top-tier AI conferences (ICML, ICCV, CVPR). Applied graph algorithms to detect research clusters, map collaboration networks, and surface emerging themes.
- Translated graph insights into interactive dashboard, enabling stakeholders to visualize local AI research landscape, identify domain experts, and prioritize strategic collaborations.

PSA International Singapore

May. 2018 - Nov. 2018

Data Science Intern, Data for Excellence department

- Utilised predictive analytics and time series forecasting techniques to forecast PSA Port Throughput with 96% R-squared. Extended into a predictive modelling dashboard using R Shiny to identify trends and dependencies due to external market uncertainties.
- Developed visualisation dashboards using PowerBI to provide actionable and analytical insights to management executives regarding PSA stakeholder analysis, competition within port business, and busiest routes for high-volume traffic routes.
- Managed the consolidation, data cleansing, and preparation of automation scripts for transshipment data sourced from PSA terminals spanning across Europe, Asia, and the Americas into an MS SQL server.



Land Transport Authority

Aug. 2015 - Dec. 2017

Executive Project Engineer (Signaling), Network Renewal Systems department

- Successfully commissioned S\$ 100M on-board train signaling systems for Circle Line and North-East Line. Co-ordinated with key stakeholders and various parties to plan on-site commissioning services within an operational facility of SMRT, SBST and executed works with appropriate recovery action plans.
- Collaborated with key stakeholders such as operations, cross-functional teams, contractors to engage in regular technical meetings, identify gaps and brainstorm solutions.
- Led warranty management of new trains for revenue service to SMRT and SBST. Joined CCL EB-EVAC task force to perform investigations on intermittent loss of signaling communications in new trains. Together with SMRT and Alstom, identified opportunities to reduce rail disruptions by 60%.
- Automated fault extraction process from data related to rail operations and delay/non-delay causing events using Excel macros, thus cutting down manual work. Presented findings to management for review and escalation, managing feedback, timelines, and action plan.
- Participated in project planning for CCL 6 through feasibility studies, preparation of tender documentation and evaluation of designer submissions.

EDUCATION

Singapore Management University

Jan. 2018 - Jun. 2019

Master of IT in Business – Analytics & AI, Distinction (GPA: 3.86/4.00)

- MITB Dean's List for outstanding academic excellence (top 5%)
- MITB Scholarship for Singapore Management University (SMU) postgraduate program.

National University of Singapore

Aug. 2011- Jul. 2015

Bachelor of Engineering (Electrical Engineering), First Class Honours

- NASA Exchange Scholarship for overseas student ambassador at Royal Institute of Technology, Stockholm, 2014.
- NUS Dean's List for outstanding academic excellence (top 5%).

SKILLS, LANGUAGES, AWARDS AND CERTIFICATES

- Certification CSPO: Certified Scrum Product Owner, Scrum Alliance, Certification ID: 1928190, Apr 2024
- Technical Skills
 - **Programming Languages:** Python, R
 - **Statistical Tools:** SAS JMP Pro, SAS Enterprise Miner
 - **Machine and Deep Learning:** Applied ML, Natural Language Processing, Generative AI, LLMs, Graph Data Science
 - **Agentic Frameworks:** LangChain, LangGraph, Defy, N8N, MCP.
 - **Data Visualisation Tools:** Tableau, PowerBI, Gephi, R Shiny
 - **Databases:** SQL, Neo4J, Elastic Search
 - **Web Framework:** Flask, FastAPI
 - **Big Data Tools:** PySpark, HiveQL, Snowflake, Google BigQuery, Apache Airflow
 - **DevOps:** Git, Jira, Jenkins, Docker, Circle CI
 - **Cloud Tools:**
 - o AWS: S3, EC2, ECR, ECS, Transcribe, lambda, API Gateway
 - o Azure: Azure OpenAI, Azure AI Foundry, MAG, AzureML, AI
 - o GCP: Dataflow, Dataproc, BigQuery, CloudSQL, Cloud Data Fusion
 - o Alibaba Cloud: DSW, MaxCompute, PAI, OSS
- Languages: English, Tamil.